Co-Boosting++: Coupled Optimization of Data and Ensemble for One-Shot Federated Learning

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***Abstract*—One-shot Federated Learning (OFL) has emerged as a promising paradigm, enabling global model training with minimal communication overhead. In OFL, the server model is usually distilled from an ensemble of pre-trained client models, while the ensemble also facilitates synthetic data generation for the knowledge distillation process. Prior works show that the performance of the final model is fundamentally tied to both the quality of the synthetic data and the ensemble. However, existing methods often optimize these two components separately, overlooking their interaction. To address this coupled optimization problem and provide a unified solution to the dual challenges of data and model heterogeneity inherent in OFL, we introduce Co-Boosting++, a novel OFL framework where synthetic data generation and ensemble construction mutually enhance each other in an iterative fashion. First, we fix the ensemble and generate hard samples in an adversarial manner. These samples are crucial for enhancing the robustness of knowledge transfer, as they challenge the model to generalize better, thereby improving quality of the synthetic data and subsequent distillation process. Second, leveraging these hard samples, we enhance the ensemble via a Mixture of Experts (MoE) mechanism. MoE allows dynamic adjustment of ensemble weights based on the generated hard samples, which enables the ensemble to better capture diverse and heterogeneous knowledge from client models. Furthermore, we extend Co-Boosting++ to support the simultaneous generation of multiple heterogeneous target models, enabling efficient adaptation to diverse device constraints. Extensive experiments on benchmark datasets demonstrate that Co-Boosting++ consistently outperforms state-of-the-art methods due to its coupled optimization of data and ensemble quality. Additionally, Co-Boosting++ is highly practical in real-world model market scenarios, requiring no local training modifications, additional transmissions, or restrictions on client model architectures.**

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Our code is available at https://github.com/rong-dai/Co-Boosting-PP.

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I. Introduction

FEDERATED Learning (FL) [1] has become a widely adopted distributed learning paradigm that enables collaborative model training across multiple clients without directly sharing their local datasets. While the traditional multi-round parameter-server communication framework facilitates effective information exchange between clients and the central server, it poses several real-world challenges. These include: 1) substantial communication overhead and the risk of unstable connections between clients and the server [2], [3], [4], [5], [6], and 2) heightened vulnerability to adversarial threats such as man-in-the-middle attacks [7] and broader privacy and security risks [8], [9], [10], [11].

To alleviate these issues, One-Shot Federated Learning (OFL) [12] has been introduced, which reduces communication rounds to a single iteration, mitigating communication failures and enhancing security by minimizing data transmission exposure. Additionally, OFL aligns well with contemporary model market scenarios [13], where clients primarily contribute pre-trained models rather than engaging in iterative training. In OFL, the target model is constructed through a data-free knowledge distillation process, where the ensemble, formed from the predictions of each client model, acts as the teacher, and synthetic data facilitates knowledge transfer from the teacher to the target model. The ensemble not only serves as a knowledge aggregator but also plays a crucial role in generating high-quality synthetic data for distillation. As demonstrated in [12] and [14], the performance of the final target model heavily depends on both the quality of the ensemble and the generated synthetic data. Thus, the primary challenge in improving performance lies in the process of improving the data and the ensemble.

Existing approaches typically address this challenge by focusing on either improving the ensemble quality or enhancing the synthetic data quality, but rarely both. To refine the ensemble, prior works such as [15], [16], and [17] modify the local training phase, often requiring additional transmissions, which may not be feasible in real-world federated learning settings. [18] employs a mixture of experts but omits the distillation step, limiting its applicability in resource-constrained edge settings. On the other hand, to improve synthetic data, methods like [19] leverage auxiliary public datasets, [20] propose transmitting distilled datasets to the server, [21] introduce auxiliary diffusion models,

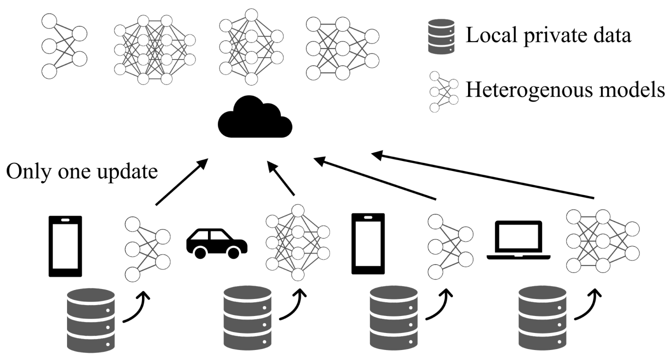

Fig. 1. The heterogeneous one-shot federated learning setting. The clients with varying computing capabilities deploy heterogeneous local models. Each client communicates with the central server only once. After fully training their local models on non-IID private data, clients send these models to the server, where they are aggregated into multiple heterogeneous models.

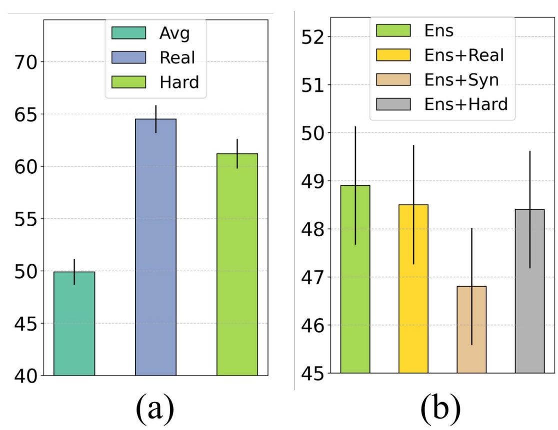

Fig. 2. Experimental comparison. (a) shows the test accuracy of ensemble with **averaged** weights, learned weights from the MoE structure using the **real** data, and learned weights from the MoE structure using the **hard** samples. (b) shows test accuracy of server obtained through distillation on **real**, **synthetic**, and **hard** samples with same ensemble. Experiments are done on CIFAR-10 with 10-client $Dir(0.1)$-parted setting.

while [14] employ data-free knowledge distillation techniques to synthesize data directly from an averaged ensemble. However, these methods typically follow a sequential approach, where improvements in either data generation or ensemble construction occur independently, overlooking the critical interaction between the two in the knowledge distillation process.

Furthermore, in practical model market scenarios, where only well-trained models with diverse architectures and trained on different data distributions are available, modifications to local training, additional data transmissions, or extra model updates are often impractical or even prohibited. This becomes even more challenging when considering device heterogeneity, as different end devices have varying computational resources and deployment constraints. Therefore, as illustrated in Fig. 1, in the OFL process, it is essential to generate heterogeneous target models tailored to different device capabilities. This raises a crucial question: how can we effectively aggregate knowledge from heterogeneous pre-trained models, through data-free knowledge distillation to obtain diverse, device-specific models? In this context, optimizing both ensemble quality and synthetic data generation during the data-free knowledge distillation process becomes a fundamental challenge that needs to be addressed.

To address these challenges, we propose Co-Boosting++, a novel one-shot federated learning algorithm, as illustrated in Fig. 3, where the synthesized data and the ensemble model mutually enhance each other in an iterative manner. For improving data quality, given a fixed ensemble model, we employ hard sample generation techniques in an adversarial fashion, ensuring that the generated samples are difficult for the student model to learn during knowledge distillation. This forces the student to extract more meaningful information, leading to better knowledge transfer. Furthermore, since the ultimate goal is to generate multiple heterogeneous target models, we introduce diversity enhancement, ensuring that the generated samples account for the learning differences among various heterogeneous models. For improving the ensemble model, we adopt a Mixture of Experts (MoE) [22] framework, where each local pre-trained model is treated as an expert, and a gating network dynamically assigns experts to different input samples. This flexible expert selection mechanism prevents the limitations of a one-size-fits-all model, effectively leveraging knowledge from diverse local models. Using the generated hard samples, we train this MoE-based ensemble, allowing it to adaptively learn expert assignments and further refine its predictive capabilities. By iteratively improving both data quality and ensemble quality, we create a coupling effect, where the enhanced data generation process supports a stronger ensemble model, and in turn, the improved ensemble further refines the data synthesis. This progressive co-boosting mechanism leads to superior model performance as can be seen in Fig. 2(a) and (b).

Thorough experiments on multiple benchmark datasets demonstrate the superiority of the proposed Co-Boosting++ algorithm. Additionally, due to its inherent design, Co-Boosting++ is highly practical in today's model market scenarios, where one-shot federated learning systems may adapt to heterogeneous client architectures without requiring local training adjustments. In summary:

- We show that it is possible to simultaneously improve both the synthetic data quality and the ensemble model, two crucial elements in one-shot federated learning. This finding paves the way for future advancements in optimizing the interaction between these components.
- We introduce Co-Boosting++, a novel one-shot federated learning method that integrates an adversarial paradigm with a MoE framework. Through an iterative process that simultaneously enhances both synthetic data and ensemble quality, Co-Boosting++ establishes a dynamic feedback loop, where improvements in data generation continuously strengthen the ensemble, and vice versa.
- We highlight Co-Boosting++'s practicality in real-world model market scenarios. It requires no local training adjustments, extra data/model transmissions, and supports diverse target model architectures.
- Extensive experiments validate Co-Boosting++'s robustness and scalability, showing it consistently outperforms state-of-the-art methods due to improved synthetic data and ensemble quality.

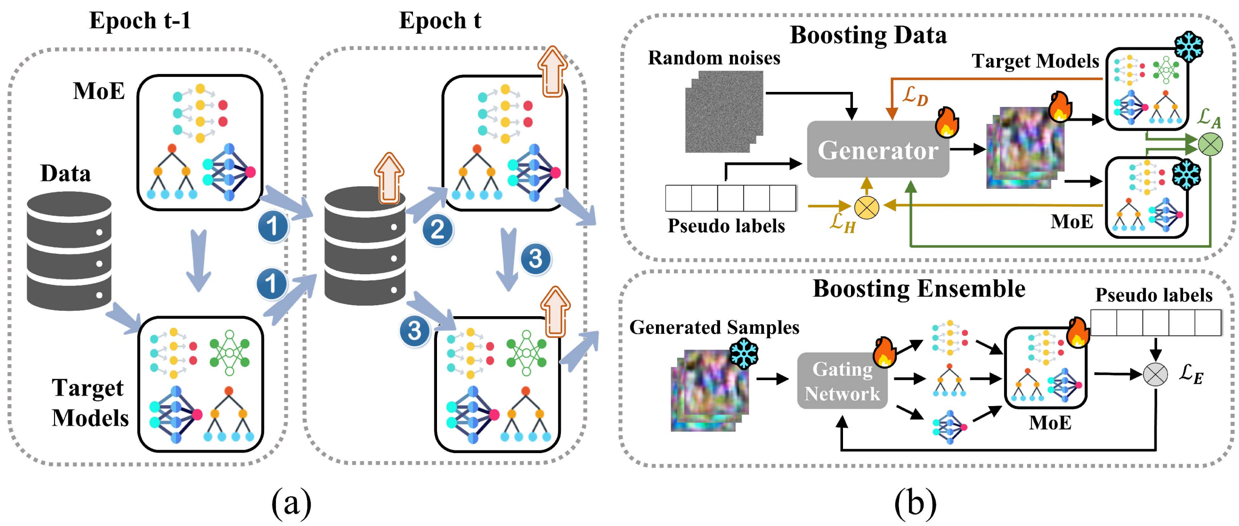

Fig. 3. Co-Boosting++ framework. (a) In each epoch, high-quality samples are first generated based on the last epoch's MoE ensemble and server target models, which are then used to adjust MoE client weights giving a better ensemble. Based on the enriched data and refined ensemble, target server models are updated by distilling knowledge from them. (b) Boosting data by generating samples through hard sample generation loss ($\mathcal{L}_H$ and $\mathcal{L}_A$) combined with a diversity-enhancing loss ($\mathcal{L}_D$). Boosting ensemble by using the generated samples to update the gating network in the MoE structure ($\mathcal{L}_E$).

In this journal version, we introduce several significant improvements over the conference version, which was accepted to ICLR 2024 [23]. First, we expand the target model generation process to consider multiple heterogeneous target models. By leveraging the modified data-free knowledge distillation approach, we now simultaneously generate models with different architectures, tailored to the requirements of diverse devices. This enhances both the diversity and adaptability of the generated models. Second, we introduce a diversity-enhanced loss function in the sample generation process. This ensures that the synthetic data captures a broader range of learning patterns, further improving the performance of models trained on the generated data. Finally, we replace the static ensemble weights used in the conference version with a Mixture of Experts (MoE) framework. This dynamic approach allows the ensemble to flexibly select experts based on the input data, improving the overall accuracy and robustness of the ensemble model. We conduct extensive experiments and comprehensive analysis involving both heterogeneous and homogeneous models, as well as scenarios with heterogeneous data, providing detailed comparisons with SOTA methods.

II. Related Works

A. Federated Learning

With the growing concern of data privacy, federated learning (FL) [1], [24], [25] allows multiple decentralized clients to collectively train a machine learning model without the need to gather all clients' data. Most FL algorithms follow the communication prototype of FedAvg [1], which requires many communication rounds to train an effective global. To minimize communication costs and mitigate potential security risks associated with multi-round communications, one-shot federated learning (OFL) has emerged as a promising research direction.

Compared to the straightforward baseline of parameter averaging, as in FedAvg [1], [12] is the first to propose OFL and utilizes the ensemble of each client's pre-trained models as the ensemble teacher. Subsequently, an auxiliary dataset is employed to distill knowledge from this ensemble and consolidate it into a unified server model. This paradigm, which is the mainstream of OFL, inherently relates the performance of the server to the data and ensemble used in the knowledge distillation stage.

With the purpose of using high-quality data, [19] proposes a two-tier knowledge distillation stage with the public dataset. Instead of using public data, [20] proposes to distill the local dataset on the client side and then send it to the server, which may help the aggregation. [21] proposes to transmit class prototype and utilize auxiliary pre-trained diffusion models. To avoid transmitting additional information, [14] proposes to generate fake data sourced from the direct ensemble and then use these fake data to distill the server model.

Regarding the improvement of the ensemble, [15] reforms the OFL task by utilizing a cluster-based method and requires uploading the cluster means. [17] modifies the local training phase of each client by introducing the placeholders in each client model. [16] alters the local model into conditional variation auto-encoders and uses these to generate samples for ensemble and knowledge distillation. [18] propose using the mixture of experts technique but omit the distillation part, resulting in an ensemble that requires storing all pre-trained models, which limits its applicability in resource-constrained edge settings.

However, none of the aforementioned approaches simultaneously enhance both the data and the ensemble quality. Furthermore, very few methods are directly applicable in real-world scenarios, particularly in modern model-market contexts [13], where only pre-trained models are provided to the server. This introduces constraints, such as the inability to alter clients' local training processes, avoid additional data transmissions, and handle client model heterogeneity. We are the first to propose a method that simultaneously improves both data and ensemble quality, and our approach is particularly well-suited for contemporary model-market scenarios, where heterogeneous models tailored to various device constraints can be effectively generated.

B. Knowledge Distillation

Knowledge Distillation (KD) [26] is a technique designed to transfer knowledge from one or more teacher networks to student models, typically by minimizing the discrepancy between their outputs. In the context of federated learning, KD enables the transfer of knowledge from multiple local clients to the global server models, enhancing the server's performance without directly sharing raw data. The concept was first introduced in federated learning by [27], who applied KD at the server side using an unlabeled auxiliary dataset.

To reduce dependence on proxy datasets, recent research has focused on data-free knowledge distillation methods. In centralized settings, techniques such as those proposed by [28] and [29] use generators or noise to simulate the teacher model's behavior, creating synthetic data for distillation. Building on this, [30] and [31] employed locally updated generators, which are aggregated globally to synthesize distillation samples.

Further developments include [32], which enhanced ensemble distillation by incorporating weighted averaging based on locally trained discriminators. However, these approaches are not ideal for OFL, where the multiple rounds of training or transmission of generators and discriminators would be impractical. Additionally, the requirement for an extra local client component contradicts the constraints in modern model-market OFL settings, which focus on using pre-trained models with limited communication overhead.

A significant challenge with locally trained generators is their potential for privacy leakage. Since the generator has direct access to the client's local private data, it could memorize and inadvertently reveal sensitive information [33]. In contrast, OFL techniques ensure that the generator is trained without access to raw data, thereby preserving privacy and adhering to the strict requirements of modern federated learning environments.

III. Methodology

In this section, we first provide an overview of the general process behind one-shot federated learning (OFL) in Section III-A, where the final performance of the server model heavily depends on both the quality of the ensemble and the synthetic data. Following this, we present our proposed method, Co-Boosting++, which innovatively addresses the data quality issue by generating high-quality synthetic data and improving overall model performance using a specialized ensemble strategy. Co-Boosting++ builds on our previous work accepted to ICLR 2024 [23], and introduces several key enhancements to further optimize the synergy between data generation and ensemble learning.

We then delve into the specifics of Co-Boosting++, detailing how we generate high-quality data through adversarial and diversity-based augmentation techniques in Section III-B. These methods ensure that the synthetic data closely mirrors real-world data distributions, thereby enhancing the robustness and generalization capabilities of the model. This is an extension of our original approach, where we now incorporate a diversity-enhanced loss function to better capture a range of learning patterns, further improving model performance.

Simultaneously, we discuss the enhancement of the ensemble model, leveraging the Mixture of Experts (MoE) architecture to refine ensemble performance in Section III-C. This dynamic approach replaces the static ensemble weights used in the conference version, allowing specialized models to collaborate more effectively. The MoE framework enables the ensemble to flexibly select experts based on the input data, boosting the overall predictive power of the system. These upgrades provide a significant improvement over the conference version by enhancing adaptability to diverse device constraints and heterogeneous data environments.

Finally, we explain how these two components—data generation and ensemble optimization—are interlinked to mutually enhance each other in Section III-D. As illustrated in Fig. 3(a), we highlight the self-boosting mechanism in which high-quality data feeds into the ensemble to refine it further, while the improved ensemble contributes to better synthetic data generation. This dynamic synergy, extended from the ICLR 2024 framework [23], forms the foundation for Co-Boosting++ and ensures effective one-shot federated learning.

A. One-Shot Federated Learning

Suppose we have a set of clients $\mathbb{C}$ with $n = |\mathbb{C}|$ clients in total. Each client $c_k \in \mathbb{C}$ possesses a pre-trained model with parameters $\boldsymbol{\theta}_C^k$, which has been trained on its local private dataset $\mathbb{D}^k = \{(\mathbf{x}_i, y_i)\}_{i=1}^{n_k}$, where $n_k = |\mathbb{D}^k|$ denotes the number of local data samples $\mathbf{x}_i$ and their corresponding labels y_i. To accommodate modern model-market scenarios and computation-restricted edge device settings, OFL aims to train multiple models with different parameter sets $\boldsymbol{\theta}_S^j$ $(j = 1, \ldots, m)$ over the aggregated dataset $\mathbb{D} \triangleq \bigcup_{k=1}^{n} \mathbb{D}^k$, leveraging the pre-trained, frozen models from each client. This process can be formulated as follows:

$$\min_{\boldsymbol{\theta}_S^1,\ldots,\boldsymbol{\theta}_S^m} \sum_{j=1}^{m} \mathcal{L}(\boldsymbol{\theta}_S^j) \triangleq \sum_{j=1}^{m} \frac{1}{|\mathbb{D}^j|} \sum_{\{\mathbf{x}_i, y_i\} \in \mathbb{D}^j} \ell_{CE}\big(f\big(\mathbf{x}_i; \boldsymbol{\theta}_S^j\big), y_i\big), \tag{1}$$

where m is the number of target models, $\ell_{CE}(\cdot,\cdot)$ denotes the cross-entropy loss function, and $f(\mathbf{x}_i; \boldsymbol{\theta}_S^j)$ denotes the prediction function of the j-th server model, which outputs the logits for input $\mathbf{x}_i$ given parameters $\boldsymbol{\theta}_S^j$.

Noticeably, in OFL, the original training set $\mathbb{D}^k$ is inaccessible, and only well-pretrained client models, parameterized by $\boldsymbol{\theta}_C^k$, are available. To address this issue, existing approaches typically employ data-free knowledge distillation (DFKD). A common strategy involves constructing an ensemble model by aggregating the pre-trained client models, which then serves as the teacher for knowledge transfer. Since real data cannot be used, synthetic data are often generated to facilitate distillation.

Here we define the ensemble model aggregated from each pre-trained client model as $A(\mathbf{x}; \{\boldsymbol{\theta}_C^k\}_{k=1}^n)$. For simplicity, we use $A(\mathbf{x})$ to denote $A(\mathbf{x}; \{\boldsymbol{\theta}_C^k\}_{k=1}^n)$ in the rest of the paper, which means the output logits of the ensemble given $\mathbf{x}$.

Given the ensemble model A, synthetic data $\mathbb{D}_S$ are generated based on its output. In particular, given a random noise $\boldsymbol{z}$ sampled from a standard Gaussian distribution and a random uniformly sampled label y_s, the generator $G(\cdot)$ with θ_G is responsible for generating the data $\mathbf{x}_S = G(\boldsymbol{z})$, forming the synthetic dataset $\mathbb{D}_S$. Typically, to make sure the synthetic data can be classified correctly with a high probability by the ensemble A, the following loss is adopted:

$$\mathcal{L}(\boldsymbol{\theta}_G) \triangleq \frac{1}{|\mathbb{D}_S|} \sum_{\{\mathbf{x}_s, y_s\} \in \mathbb{D}_S} \ell_{CE}(A(\mathbf{x}_s), y_s). \tag{2}$$

After obtaining the synthetic dataset $\mathbb{D}_S$ using the generator defined in (2), OFL aims to distill knowledge from the ensemble model A into multiple models $\{\boldsymbol{\theta}_S^j\}_{j=1}^m$ with the help of these synthetic samples, which is formulated as:

$$\min_{\{\boldsymbol{\theta}_S^j\}_{j=1}^m} \sum_{j=1}^{m} \mathcal{L}(\boldsymbol{\theta}_S^j) \triangleq \frac{1}{|\mathbb{D}_S|} \sum_{\mathbf{x}_s \in \mathbb{D}_S} \sum_{j=1}^{m} \ell_{KL}(A(\mathbf{x}_s), f(\mathbf{x}_s; \boldsymbol{\theta}_S^j)), \tag{3}$$

where $\ell_{KL}(\cdot,\cdot)$ denotes the Kullback-Leibler (KL) divergence, measuring the discrepancy between the ensemble's output $A(\mathbf{x}_s)$ and the predictions of each server model $f(\mathbf{x}_s; \boldsymbol{\theta}_S^j)$. This formulation allows the framework to generate multiple heterogeneous models, making it adaptable to scenarios with diverse computational and deployment constraints.

If only a single target model is needed, the formulation simplifies by selecting a single instance of $\boldsymbol{\theta}_S^j$ and removing the summation over m, thereby reducing (1) and (3) to the standard single-model distillation objective.

Existing works illustrate that the performance of the server model is intrinsically related to the synthetic data $\mathbb{D}_S$ and the ensemble A, which can also be concluded according to (3).

B. Boosting Data Quality

Synthesizing data $\mathbb{D}_S$ is used to distill the ensemble model into the final server models as in (3). The quality of these synthesized data has been demonstrated vital to the distillation stage [27]. Moreover, since these data are also generated from the ensemble as in (2), it is of great importance to make these data embed as much of the knowledge of the ensemble as possible and make them transferable to the final server models.

However, as hinted in [34] and [14], by utilizing only the CE loss, the synthesized data can be easily fitted by the server model, resulting in poor performance in the knowledge distillation stage. Therefore, to improve the quality of the generated data and make them focus more on transferable components, taking inspiration from [35] and [36], we increase the importance of hard samples while suppressing the importance of easy-to-fit samples in the generation stage. More specifically, given a prediction function f which outputs logits, we employ the GHM introduced in [37] to measure the sample difficulty d of $\mathbf{x}$:

$$d(\mathbf{x}, f) = 1 - \sigma(f(\mathbf{x}; \boldsymbol{\theta}))_y, \tag{4}$$

where $\sigma(f(\mathbf{x}; \boldsymbol{\theta}))_y$ is the probability on label y predicted by the function $f(\cdot)$ with $\boldsymbol{\theta}$. Built upon the sample difficulty, we propose a hard-sample-enhanced loss $\mathcal{L}_H$ to synthesize data:

$$\mathcal{L}_H(\mathbf{x}_s, y_s; \boldsymbol{\theta}_G) \triangleq \frac{1}{|\mathbb{D}_S|} \sum_{\{\mathbf{x}_s, y_s\} \in \mathbb{D}_S} d(\mathbf{x}_s, A) \ell_{CE}(A(\mathbf{x}_s), y_s). \tag{5}$$

Moreover, to make synthesized samples hard for the server model to fit, an adversarial loss [38] is also introduced to generate hard samples. We try to maximize the differences in predictions between the ensemble model and the server models when generating data as follows:

$$\mathcal{L}_A(\mathbf{x}_s, \boldsymbol{\theta}_S^j; \boldsymbol{\theta}_G) \triangleq \frac{1}{|\mathbb{D}_S|} \sum_{\{\mathbf{x}_s, y_s\} \in \mathbb{D}_S} -\ell_{KL}\big(A(\mathbf{x}_s), f\big(\mathbf{x}_s; \boldsymbol{\theta}_S^j\big)\big). \tag{6}$$

When multiple models need to be trained, it is crucial to ensure that the synthetic samples effectively capture diverse knowledge relevant to different target models. This scenario can be regarded as a multi-teacher multi-student knowledge distillation problem. To further improve the quality of synthetic samples, we propose a sample diversification enhancement loss, which encourages the generated samples to exhibit significant variations across different target models, as in:

$$\mathcal{L}_D(\mathbf{x}_s; \boldsymbol{\theta}_G) \triangleq \frac{1}{|\mathbb{D}_S|} \sum_{\mathbf{x}_s \in \mathbb{D}_S} \sum_{i<j} d\left(f(\mathbf{x}_s; \boldsymbol{\theta}_S^i), f\left(\mathbf{x}_s; \boldsymbol{\theta}_S^j\right)\right), \tag{7}$$

where $f(\mathbf{x}_s; \boldsymbol{\theta}_S^i)$ and $f(\mathbf{x}_s; \boldsymbol{\theta}_S^j)$ are the predictions of two different target models, and $d(\cdot,\cdot)$ represents a discrepancy measure, using KL divergence to encourage diversity among the learned outputs.

By combining the above losses, we can obtain the loss used to train the generator as follows:

$$\mathcal{L}(\boldsymbol{\theta}_G) \triangleq \mathcal{L}_H + \alpha \frac{1}{m} \sum_{j=1}^{m} \mathcal{L}_A + \beta \mathcal{L}_D, \tag{8}$$

where α and β are the scaling factors for the losses, which are both set as 1in the implementations.

Though samples synthesized using (8) are hard to fit for the current ensemble model, their difficulties for the server models are still lacking. This stems from the fact that the server model can easily fit these limited unchanged data during multiple distillation steps. To further promote the sample difficulty and diversity on the fly, we draw inspiration from adversarial learning [39], [40] to generate hard and diverse samples for the server models to learn.

More specifically, we diversify and increase the sample difficulty $d(\mathbf{x}_s, A)$ on the fly to make the synthetic samples hard to fit by introducing a perturbation $\boldsymbol{\delta}_i$ for each $\mathbf{x}_s$:

$$\boldsymbol{\delta}_i = \arg \max_{\|\boldsymbol{\delta}'\|_\infty \le \epsilon} d\left(\mathbf{x}_s + \boldsymbol{\delta}', A\right), \tag{9}$$

where $||\cdot||_\infty$ represents the $\mathcal{L}_\infty$-norm and ϵ controls the strength of perturbation. To simplify the computation and enhance the diversity, instead of the iterative adversarial attacks, we take only one step of loss backward to seek the direction in the input space that maximizes the similarity between the model output and a randomly sampled vector. The hard samples $\tilde{\mathbf{x}}_s$ are constructed as follows:

$$\tilde{\mathbf{x}}_s \triangleq \mathbf{x}_s + \epsilon \frac{\bigtriangledown_{\mathbf{x}_s}(\boldsymbol{u}^\top A(\mathbf{x}_s))}{||\bigtriangledown_{\mathbf{x}_s}(\boldsymbol{u}^\top A(\mathbf{x}_s))||_2}, \tag{10}$$

where $\boldsymbol{u} \sim Unif([-1,1])^d$ is a randomly sampled vector with dimension d. Following these, the originally generated hard samples are further harder and more diverse due to the randomness in $\boldsymbol{u}$.

By replacing each sample x_s in $\mathbb{D}_S$ with $\tilde{\mathbf{x}}_s$, we can achieve a more hard and diverse synthetic dataset $\mathbb{D}_S$. Utilizing these hard samples, the knowledge of the ensemble model is transferred to the server models by knowledge distillation, as in (3).

Overall, with the hard sample technique embedding in the data synthesizing stage (replacing the generator loss in (2) with (8) and making them diverse in the distillation stage (reconstruct the synthetic dataset $\mathbb{D}_S$ according to (10)), the quality of data generated and used for distillation becomes better, naturally boosting the performance of the target models.

C. Boosting Ensemble Quality

The ensemble model takes the role of aggregating knowledge from all pre-trained models $\{\boldsymbol{\theta}_C^k\}_{k=1}^n$ and forms a virtually best-performing teacher. A straightforward method is to obtain the global model by averaging the parameters of all client models (e.g. FedAvg [1]). However, FedAvg may fail to deliver a good performance when data among clients are non-IID [41], [42] and cannot handle the challenge of client model heterogeneity. Recent works [16], [17] intend to construct a better ensemble model by altering the local training phase of each client, which may be unreliable, especially in today's model market scenarios. To tackle the client model heterogeneity and make the ensemble more practical, [12], [14] utilizes the direct ensemble A with $w_k = 1/n$ as the teacher, which means averaging the logits or weighted averaging them according to the number of the client data ($w_k = n_k/\sum_{k=1}^n n_k$). However, as suggested by [43] and [32], the simple averaging or weighted averaging based on the client data amount may not be effective, especially in non-IID settings. There exists a better combination of each client's contribution.

Mixture of Experts (MoE) [22] is an ensemble learning technique designed to improve overall model performance by utilizing multiple expert networks. Each expert in an MoE model is responsible for processing a specific type of data or task. The key component of MoE is the gating network, which selectively routes input data to the most relevant experts, ensuring that only the appropriate parts of the model are activated for each input. Motivated by this approach, we treat each pre-trained client model as an expert and propose incorporating MoE into our ensemble framework. This enables us to address the challenges posed by client model heterogeneity and optimize performance by effectively selecting the most suitable models for each data point.

For the pre-defined ensemble $A(\mathbf{x}; \{\boldsymbol{\theta}_C^k\}_{k=1}^n)$, we introduce a gating network $\mathcal{G}$ that dynamically adjusts the contribution of each expert, i.e., the pre-trained client models, based on the input. The gating network optimizes the influence of all experts by adaptively assigning weights, ensuring that each expert's knowledge is leveraged effectively in different scenarios. The dynamic weights are calculated as in:

$$w(\mathbf{x};\mathcal{G}) = \text{softmax}(\mathcal{G}(\mathbf{x})), \tag{11}$$

where $\mathcal{G}(\mathbf{x})$ denotes the output of the gating network given the input. For any input sample, the gating network dynamically adjusts the weights of each expert and assigns the sample to specialized experts for optimal performance, resulting in a dynamically weighted, sample-wise ensemble:

$$A\left(\mathbf{x}; \{\boldsymbol{\theta}_C^k\}_{k=1}^n, \mathcal{G}\right) \triangleq \sum_{i=1}^n w_k\left(\mathbf{x};\mathcal{G}\right) f\left(\mathbf{x};\boldsymbol{\theta}_C^i\right), \tag{12}$$

where $f(\mathbf{x};\boldsymbol{\theta}_C^i)$ are the predictions of each expert.

To this end, we propose enhancing the ensemble quality by optimizing a more effective weighted combination of logits through training the gating network. As demonstrated by our experimental results in Fig. 2, given high-quality data (validation data), we can achieve a better ensemble with weights different from simple averaging or data amount based averaging. Fortunately, instead of using auxiliary data, we actually can acquire high-quality generated data from the hard synthesized samples set $\mathbb{D}_S$.

In practice, the gating network can exhibit an over-trust behavior, where it disproportionately favors only a few experts in the ensemble, as noted in [44], [45]. We introduce a balance loss aimed at reducing this bias in the network's weight assignment. Specifically, we compute the importance of each expert for every batch at iteration t, and the balance loss is calculated as the square of the coefficient of variation (CV) of these importance values, as shown in (13). In our implementation, we set the scaling factor λ to 1, giving

$$\mathcal{L}_B(\mathbf{x};\mathcal{G}) = \lambda \cdot \text{CV}\left(\sum_{\mathbf{x}} w\left(\mathbf{x};\mathcal{G}\right)\right)^2. \tag{13}$$

This balance loss serves as a regularizer in the training process, promoting diversity in the expert usage and preventing the gating network from overly relying on a small subset of experts. The overall MoE training loss combines two components: the first focuses on optimizing performance by minimizing the discrepancy between the model outputs and the true labels, while the second aims to ensure a more balanced contribution from each expert, thus preventing over-trust, as in:

$$\mathcal{L}_E(\mathbf{x};\mathcal{G}) = \frac{1}{|\mathbb{D}_S|}\sum_{\{\mathbf{x}_s,y_s\}\in\mathbb{D}_S} \ell_{CE}(A(\mathbf{x}_s),y_s). + \mathcal{L}_B(\mathbf{x};\mathcal{G}), \tag{14}$$

where $A(\mathbf{x}_s)$ is short for $A(\mathbf{x}; \{\boldsymbol{\theta}_C^k\}_{k=1}^n, \mathcal{G})$ in (12).

The reweighting of each client's logit results in a superior ensemble model, which will naturally benefit the server model.

Algorithm 1: Co-Boosting++.

Input: Clients' local pre-trained models $\{\boldsymbol{\theta}_{\boldsymbol{C}}^{\mathbf{1}}, \ldots, \boldsymbol{\theta}_C^n\}$, target server models $\{\boldsymbol{\theta}_{\boldsymbol{S}}^{\mathbf{1}}, \ldots, \boldsymbol{\theta}_S^m\}$, synthetic dataset $\mathbb{D}_S = \emptyset$, ensemble A, generator $\boldsymbol{\theta}_G$, gating network $\mathcal{G}$, perturbation strength ϵ, learning rate of generator and server η_G and η_S, generation iterations T_G, global model training epochs T, and batch size b
Output: Target server models $\{\boldsymbol{\theta}_{\boldsymbol{S}}^{\mathbf{1}}, \ldots, \boldsymbol{\theta}_S^m\}$
for epoch $= 0$ to $T - 1$ **do**
  // Generate hard synthetic samples
  Sample a batch of noises and labels $\{\boldsymbol{z}_i, y_i\}_{i=1}^b$
  for $t_g = 0$ to $T_G - 1$ **do**
    Generate $\{\mathbf{x}_s\}_{i=1}^b$ with $\{\boldsymbol{z}_i\}_{i=1}^b$ and $\boldsymbol{\theta}_G$
    Update the generator:
    $\boldsymbol{\theta}_G \leftarrow \boldsymbol{\theta}_G - \eta_G \bigtriangledown_{\boldsymbol{\theta}_G} \mathcal{L}(\boldsymbol{\theta}_G)$, where $\mathcal{L}(\boldsymbol{\theta}_G)$ is defined in (8)
  end for
  $\mathbb{D}_S \leftarrow \mathbb{D}_S \cup \{\mathbf{x}_s\}_{i=1}^b$
  Diverse each sample $\{\mathbf{x}_s\}$ to $\{\tilde{\mathbf{x}}_s\}$ according (10)
  // Obtain a better ensemble
  Update gating network $\mathcal{G}$ with $\mathbb{D}_S$ according (14)
  // Obtain the final server models
  for sampling batch $\{\mathbf{x}_s\}$ in $\mathbb{D}_S$ **do**
    Update server models:
    $\boldsymbol{\theta}_S \leftarrow \boldsymbol{\theta}_S - \eta_S \bigtriangledown_{\boldsymbol{\theta}_S} \mathcal{L}(\boldsymbol{\theta}_S)$, where $\mathcal{L}(\boldsymbol{\theta}_S)$ is defined in (3)
  end for
end for

Moreover, since the operations are done on the logit layer, this reweighting technique can be easily applied to both heterogeneous and homogeneous client model settings.

D. Co-Boosting Data and Ensemble

As aforementioned, we introduce how to boost the data quality by utilizing hard sample techniques with a fixed ensemble and how to boost the ensemble with a fixed synthetic dataset. Actually, these two stages are inherently entangled and can boost each other at the same time. To get high-quality ensemble A and synthesized data $\mathbb{D}_S$, combining the definitions above, our proposed method can be viewed as attempting to solve the following problem:

$$\min_{\mathcal{G}} \frac{1}{|\mathbb{D}_S|} \sum_{\{\mathbf{x}_s, y_s\} \in \mathbb{D}_S} \max_{\delta \in S} \ell_{CE}(A(\mathbf{x} + \delta; \{\boldsymbol{\theta}_C^k\}_{k=1}^n, \mathcal{G}), y_s), \tag{15}$$

where y_s is the label of sample $\mathbf{x}_s$, and δ is the perturbation constrained in S. This problem can be addressed adversarially, which means the improvement of the data and the ensemble can be done simultaneously. With better quality data, the weighted ensemble can reach higher performance, while with this better ensemble, the data synthesized sourced from this ensemble can further embed more knowledge. Therefore, by mutually boosting the quality of the synthesized data and the ensemble, we can naturally get a better-performance server model through the distillation in (3).

The overall algorithm is summarized in Algorithm 1. In each iteration, we first generate hard samples based on the current ensemble model and the last epoch server models. With these generated data, an enhanced ensemble model is cultivated by searching for the optimal ensembling gating weights of each client. Utilizing the generated data and the upgraded ensemble, the final server models are trained by distilling the ensemble on these data. As illustrated in Sections III-B and III-C, with either one fixed, one can benefit from the other, thus by mutually boosting each other in the proposed Co-Boosting++, we can achieve both better quality data and the ensemble periodically. Therefore, the global server models trained on them will inherently become better than other methods.

In contrast to the ICLR 2024 version [23], which primarily focused on the interplay between synthetic data generation and a static ensemble, Co-Boosting++ introduces key extensions that allow for dynamic and iterative enhancement. Specifically, we now incorporate a diversity-enhanced loss function to improve the data generation process and replace the static ensemble weights with a Mixture of Experts (MoE) framework. These improvements allow the ensemble to flexibly adapt to the data and further refine both the data generation and model performance, making Co-Boosting++ more adaptable to diverse devices and heterogeneous data environments.

IV. Experiments

A. Experimental Details

Datasets and partitions: We conduct experiments on four real-world image classification datasets that are standard in the FL literature: FMNIST [46], SVHN [47], CIFAR10, and CIFAR100 [48]. To simulate statistical heterogeneity, we use Dirichlet distribution to generate disjoint non-IID client training datasets as in [14] and [16]. In particular, we sample $\boldsymbol{p}_k \sim Dir(\alpha)$ and allocate a $\boldsymbol{p}_k^i$ proportion of the data of class i to client k. The parameter α controls the level of statistical imbalance, with a smaller α inducing more skewed label distributions among clients.

Baselines: Within the contemporary model-market scenarios, we compare the performance of Co-Boosting++ against three existing methods: FedAvg [1], DENSE [14], and our conference version [23]. To ensure fair comparisons, we omit comparisons with methods that require the use of auxiliary public datasets, such as [19], or the modification of the local training phases of each client, as seen in [17] and [16]. Both of these approaches may not be applicable in real-world scenarios. Moreover, since there is only one communication round, standard FL algorithms based on regularization terms including FedProx [49], SCAFFOLD [41], FedGamma [50], FedLada [51] and FedDyn [42] have no effect. Furthermore, since data-free one-shot FL can be addressed simply by operating data-free KD methods on the direct ensemble model of all models, similar to [14], we also introduce two prevalent data-free KD methods DAFL [28] and ADI [29] into OFL resulting in F-DAFL and F-ADI. More

TABLE I
AVERAGED TEST ACCURACY OF FIVE HOMOGENEOUS TARGET SERVER MODELS OF DIFFERENT METHODS OVER FIVE DATASETS AND ACROSS THREE LEVELS OF STATISTICAL HETEROGENEITY (LOWER α IS MORE HETEROGENEOUS).

Method	α	FedAvg	FedDF	F-ADI	F-DAFL	DENSE	Co-Boosting [23]	Co-Boosting++
FMNIST	0.05	20.07±1.98	44.73±0.40	42.25±2.01	41.66±0.83	44.77±1.87	50.62±1.13	**53.21±1.08**
	0.1	46.61±1.94	68.40±1.80	63.19±1.99	67.81±0.93	69.43±1.94	74.86±1.70	**76.63±1.14**
	0.3	60.13±2.62	83.14±0.47	74.80±1.72	78.68±0.49	81.31±0.83	83.11±1.28	**84.76±1.07**
SVHN	0.05	39.41±2.19	60.79±0.52	56.58±1.05	59.38±1.19	60.24±1.31	65.40±0.86	**67.46±1.08**
	0.1	46.22±1.92	68.98±0.63	66.33±1.69	67.77±0.34	68.30±1.01	72.88±1.19	**74.84±1.16**
	0.3	72.61±2.06	79.78±0.55	76.75±1.47	78.01±1.02	78.73±0.84	81.31±1.09	**83.66±0.89**
CIFAR-10	0.05	17.49±2.51	37.53±0.67	36.94±1.70	37.82±1.30	38.37±1.08	47.20±0.81	**49.36±0.97**
	0.1	27.54±1.80	49.63±0.80	47.19±0.97	46.32±0.97	47.80±1.21	57.09±0.94	**60.98±1.45**
	0.3	46.39±2.37	67.18±0.60	60.60±1.32	65.89±1.69	66.77±1.55	70.24±1.56	**72.28±0.98**
CIFAR-100	0.05	6.45±0.92	16.07±0.54	13.75±1.01	15.79±0.21	16.17±1.33	19.24±1.42	**20.86±1.34**
	0.1	10.28±1.70	22.07±0.43	19.44±1.66	20.99±1.17	22.21±1.41	23.59±1.27	**25.16±1.05**
	0.3	15.22±2.08	30.71±0.53	26.14±1.37	28.79±1.25	30.33±1.24	31.30±1.30	**32.52±0.89**

specifically, F-DAFL and F-ADI aim to either optimize a generator or a noise to mimic the performance of the ensemble teacher model. The data generated from this process is then utilized to distill knowledge into the student (server) models. Additionally, we include FedDF [27] using the validation dataset for distillation.

Configurations: Unless otherwise stated, we conduct experiments with 10 clients and Dir(0.1) (highly heterogeneous) partition. Results are reported averaged across at least 3 random seeds. For each client's local training, we use SGD optimizer with momentum=0.9 and learning rate=0.01. We set batch size to 128 and local epoch to 300. Once the local clients complete their training, they are frozen to align with the OFL configuration. Following [1], we use a simple CNN with 5 layers for SVHN, CIFAR10, and CIFAR100, LeNet-5 [52] for FMNIST. All available test data is used to evaluate the target server models (or ensemble). The generator we use is the same as in [14], [28] and it is trained by Adam optimizer with a learning rate $\eta_g = 0.001$ over $T_G = 30$ rounds. We use the same gating network as in [45], with an Adam optimizer and a learning rate of $\eta_{\mathcal{G}} = 0.001$. The initial weight of the gating network is set to the average weight. The distillation temperature in the knowledge distillation stage is set to 4, while the temperature used in the KL loss in the generator loss is set to 1. The perturbation strength is set to $\epsilon = 8/255$. For the training of the server models $\boldsymbol{\theta}_S$, we use the SGD optimizer with learning rate $\eta_S = 0.01$ and momentum=0.9. The number of total epochs T is set to 500. We maintain the same hyperparameters for all baseline methods.

B. Overall Performance Comparison

1) Homogeneous Setting: To evaluate the effectiveness of our method, we conduct experiments under various non-IID settings by varying $\alpha = \{0.05, 0.1, 0.3\}$ and report the performance across different datasets and methods in Table I. We consider a homogeneous setting where the target models share the same architecture as the pre-trained ones and set the number of target server models to five. Notice, that we use the validation set in FedDF, which is not practical in reality.

From the table, we can conclude that Co-Boosting++ consistently outperforms all other baselines in all settings. Notably, in many settings, Co-Boosting++ achieves over a 5% accuracy improvement compared to the best baseline, DENSE. In cases of extreme statistical heterogeneity, such as when $\alpha = 0.05$, Co-Boosting++ surpasses the best baseline by substantial margins with 8.44% , 7.22% , 10.99% , and 4.69% on FMNIST, SVHN, CIFAR-10, and CIFAR-100, respectively. We also compare the performance of the ensemble (i.e., the teacher used for knowledge distillation) across different methods on SVHN and CIFAR-10, as shown in Table II. FedENS denotes averaged ensemble. The superior performance of the server model can be attributed to the improved quality of synthesized data using hard samples and the ensemble with the MoE design. While all compared methods, except FedAvg, utilize the direct logits averaging ensemble as the ensemble and aim to distill knowledge from it in a data-free manner, Co-Boosting++, with its co-enhancing technique, results in a superior ensemble teacher, surpassing FedENS significantly. The improvement of Co-Boosting++ over the previous version Co-Boosting can be attributed to its enhanced ensemble and diversity designs, which enrich knowledge transfer. In a word, the superiority of our proposed method can be attributed to the enhanced data and ensemble quality, which naturally translates into better server models.

2) Heterogeneous Setting: To evaluate our method in a heterogeneous client setting, we apply five different models on the client side under a CIFAR-10, $Dir(0.1)$-partitioned setting. The heterogeneous models include CNN1 from [1], CNN2 from the PyTorch tutorial [53], ResNet [54], MobileNet [55], and ShuffleNet [56]. On the target server side, we use the same set of models: CNN1, CNN2, ResNet, MobileNet, and ShuffleNet. This setup aims to transfer knowledge from pre-trained models with heterogeneous architectures to target server models with different architectures, reflecting real-world scenarios where OFL must aggregate knowledge from diverse networks and deploy it to heterogeneous edge devices.

Table III presents results of comparable methods, where Local denotes directly using the pre-trained model for testing. We omit FedAvg as it does not support this setting. All methods share

TABLE II
TEST ACCURACY OF THE ENSEMBLE IN SVHN, AND CIFAR-10IN HOMOGENEOUS Dir-PARTED SETTING

Dataset	SVHN			CIFAR-10		
Method	α=0.05	α=0.1	α=0.3	α=0.05	α=0.1	α=0.3
FedENS	61.62±1.61	69.71±0.68	80.54±0.57	42.34±0.67	49.99±0.85	69.61±0.50
Co-Boosting [23]	65.69±1.48	73.52±1.71	82.90±1.35	48.75±1.25	59.86±1.76	72.67±1.27
Co-Boosting++	**69.52±0.86**	**76.31±0.85**	**86.11±1.41**	**52.50±1.33**	**62.79±1.24**	**74.74±1.44**

TABLE III
TEST ACCURACY OF THE TARGET SERVER MODEL WITH HETEROGENEOUS ARCHITECTURES ON CIFAR-10 ACROSS THREE LEVELS OF STATISTICAL HETEROGENEITY IN THE HETEROGENEOUS SETTING.

Target	$Dir(\cdot)$	Local	FedDF	F-ADI	F-DAFL	DENSE	Co-Boosting [23]	Co-Boosting++
CNN1	α=0.05	41.06±1.49	52.96±1.66	51.69±1.39	51.73±0.90	52.66±1.43	56.10±1.64	**57.28±0.94**
	α=0.1	48.05±1.68	57.66±0.76	56.18±1.67	57.04±1.54	58.31±1.21	61.87±1.86	**63.11±0.83**
	α=0.3	60.16±0.75	67.08±0.86	68.94±0.92	67.55±1.22	67.89±1.08	70.06±0.92	**72.08±1.15**
CNN2	α=0.05	30.03±0.69	44.09±0.51	43.38±1.73	43.13±1.61	43.82±1.49	45.82±1.36	**48.54±1.26**
	α=0.1	39.90±0.25	46.21±1.36	43.03±1.56	45.56±1.94	45.88±2.32	48.31±1.60	**49.88±1.22**
	α=0.3	51.79±0.75	58.52±1.41	57.49±0.86	57.43±1.19	58.15±1.53	62.45±1.58	**65.08±1.21**
ResNet	α=0.05	46.49±1.97	56.51±1.87	55.64±1.96	55.14±1.51	55.79±1.75	58.64±1.83	**60.01±1.29**
	α=0.1	51.70±1.14	59.49±1.06	60.30±1.70	58.58±1.49	58.67±0.88	62.30±0.90	**63.44±1.35**
	α=0.3	67.61±1.54	70.53±1.19	71.61±1.24	71.92±1.95	72.98±1.61	75.02±1.37	**76.06±1.09**
MobileNet	α=0.05	37.32±1.52	49.47±1.93	50.55±1.31	49.48±0.66	50.35±1.33	52.97±1.18	**55.00±1.10**
	α=0.1	44.72±1.39	50.25±1.04	49.36±1.44	51.91±0.98	51.47±0.52	54.75±1.58	**57.04±1.06**
	α=0.3	55.54±1.31	60.60±0.38	60.42±1.07	60.11±1.12	61.18±0.84	64.20±1.07	**66.42±1.46**
ShuffleNet	α=0.05	38.09±1.77	51.05±0.23	48.75±0.68	49.95±1.17	50.01±1.98	53.45±1.93	**55.45±1.13**
	α=0.1	47.12±1.71	53.15±1.68	52.88±1.41	52.04±1.36	52.25±1.00	55.72±0.77	**56.96±0.87**
	α=0.3	55.49±0.80	61.82±0.89	60.30±0.73	59.98±1.93	61.19±1.24	63.22±1.19	**64.32±1.28**

TABLE IV
TEST ACCURACY OF THE TARGET SERVER MODEL OF DIFFERENT METHODS IN CIFAR-10 UNDER C_cls-PARTITIONED SETTING

C_cls	FedAvg	FedDF	F-ADI	F-DAFL	DENSE	Co-Boosting [23]	Co-Boosting++
2	16.15±2.61	23.07±0.83	23.16±1.12	24.53±0.57	23.85±0.93	36.37±1.85	**38.05±0.90**
3	26.47±1.46	38.39±0.64	36.06±1.93	38.13±1.17	38.14±1.38	53.91±1.80	**56.76±0.90**
4	33.78±2.42	54.51±1.12	51.04±1.40	52.53±0.97	51.53±1.79	58.00±1.59	**61.04±1.43**
5	35.95±1.96	58.34±1.58	55.27±2.06	54.67±1.26	56.79±1.03	62.52±1.75	**65.86±1.23**

the same optimization hyperparameters across model architectures. As noted in [14] and [57], FL under both non-IID data and heterogeneous model architectures is highly challenging. Even in this setting, our proposed Co-Boosting++ consistently outperforms baselines by a large margin, benefiting from jointly improving ensemble learning and data synthesis. While it is natural to observe variations in performance among different architectures, this can be attributed to the use of the same hyperparameters across all architectures and the inherent differences in representation capacity among them. However, it is noteworthy that Co-Boosting consistently outperforms other baselines regardless of the server architecture. This further underscores the effectiveness of the proposed Co-Boosting++ technique in enhancing both data and ensemble learning. Additionally, the diversity loss in hard sample synthesis enriches the transferred information, contributing to the performance improvement.

C. In-Depth Study

1) Effects of Each Component: We study the effectiveness of our hard sample generation loss (8) in Section III-B, on-the-fly sample difficulty promotion (10) in Section III-B, and ensemble enhancing in Section III-C. Table V shows the experimental results on SVHN and CIFAR-10 in a 10-client $Dir(0.05)$ parted setting. Table VI confirms that the dual-loss strategy is essential for effective hard sample generation. Results in Table V illustrate that individually improving either data or ensemble leads to noticeable enhancements in the final server model performance. However, the most remarkable results are achieved

TABLE V
ABLATIONS ON DIFFERENT COMPONENTS OF OUR METHOD. "GHS" FOR HARD SAMPLE GENERATION IN THE GENERATOR LOSS, "DHS" FOR ON-THE-FLY DIVERSE HARD SAMPLE CREATION, AND "EE" FOR ENSEMBLE ENHANCEMENT THROUGH MOE REWEIGHTING.

GHS	DHS	EE	SVHN	CIFAR-10
			58.46	39.72
✔			62.24	44.25
	✔		61.38	43.75
		✔	64.67	44.45
	✔	✔	64.03	45.96
✔	✔		65.49	47.36
✔		✔	65.50	47.61
✔	✔	✔	**67.46**	**49.36**

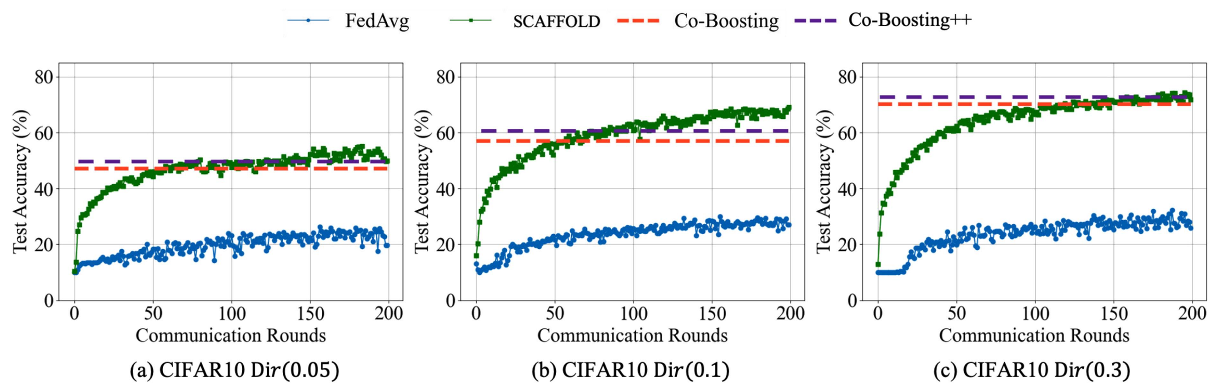

Fig. 4. Performance comparison with multi-round federated learning methods.

TABLE VI
ABLATIONS ON DIFFERENT LOSSES IN GHS

$\mathcal{L}_A$	$\mathcal{L}_D$	SVHN	CIFAR-10
✔		60.02	41.80
	✔	59.66	42.15
✔	✔	62.24	44.25

TABLE VII
COMPARISON OF TOTAL TRAINING TIME (S) AND ACCURACY (%) OF DIFFERENT METHODS USING A FIXED NUMBER OF TRAINING EPOCHS ON CIFAR-10

Methods	F-ADI	F-DAFL	DENSE	Co-Boosting	Co-Boosting++
Time	9,227	11,882	13,222	16,503	18,045
Accuracy	47.19	46.32	47.80	57.09	60.98

when both data quality and ensemble capability are improved simultaneously. This finding strongly aligns with the underlying motivation of our study.

2) Discussion on Computational Efficiency: A comprehensive evaluation of Co-Boosting++ must account for computational complexity. We emphasize that the computational overhead of our framework is strictly confined to the server-side. This cost arises primarily from the iterative training of two auxiliary components: the Generator (G) for synthetic data and the MoE gating network for ensemble enhancement. This structure represents a deliberate trade-off inherent to the OFL paradigm, where we leverage centralized server computation to achieve zero client intervention and minimal communication overhead, the most critical resource in real-world model-market scenarios. To validate the efficiency of this server-side investment, we benchmark Co-Boosting++ against powerful multi-round federated learning algorithms. As demonstrated in Fig. 4, our single-round framework successfully surpasses the performance of multi-round FedAvg and remains highly competitive with the complex multi-round SCAFFOLD algorithm, despite SCAFFOLD incurring hundreds of additional communication rounds and substantial client-side overhead. This compelling comparative analysis confirms that the server-side computational investment of Co-Boosting++ is highly justified, yielding superior overall efficiency and performance under the minimal communication budget required by practical OFL applications. We also compare the total running time and accuracy of Co-Boosting++ and various baselines in Table VII. Although our method requires more computation due to the coupled optimization of data and ensemble, it is a one-time server-side cost that is well-justified by the significant performance gains in the one-shot FL setting.

3) Different Local Data Partition: Following [17], we also conduct experiments on a C_cls-partitioned setting, which means each client only possesses data of C out of all classes. Results in Table IV further demonstrate the superiority of our proposed method. With better data quality and better ensemble, our method consistently achieves the best server model.

4) Federated Domain Generalization: It is also of great importance to evaluate our proposed Co-Boosting under scenarios involving feature shifts. To this end, we extend our experiments to include domain generalization datasets, specifically PACS [58]. Experiments follow the settings outlined in [43], [59], [60], which focus on federated domain generalization. In these experiments, each domain is allocated to a different client, and we employ the leave-one-domain-out testing technique. We utilize ResNet18 for pre-trained models. To further evaluate our method's versatility, we tested with both CNN and ResNet18 as target server architectures. Results in Table VIII demonstrate the superior performance of our proposed method, which we attribute to the mutual enhancement principle inherent in our approach. The results not only validate our method's model-agnostic feature but also its effectiveness in settings with feature distribution shifts.

5) Combination With Advanced Local Training: To alleviate the non-IID data problem in FL, recent works [50], [61], [62] have shown that the use of the Sharpness Aware Minimization (SAM) technique can be beneficial. It is suggested that incorporating this technique into local training can lead to improved server aggregation. In this section, to investigate the performance of methods utilizing SAM for local training in the one-shot federated learning setup, we conduct experiments on SVHN and CIFAR-10 datasets. The experiments are done with 10 clients under a Dir(0.05) partition scenario. From Table IX, it is evident

TABLE VIII
TEST ACCURACY OF THE TARGET SERVER MODEL OF DIFFERENT METHODS IN PACS DATASET OVER TWO ARCHITECTURES ON THE UNSEEN DOMAIN USING LEAVE-ONE-OUT TEST MECHANISM

Server	Domain	FedAvg	FedDF	F-ADI	F-DAFL	DENSE	Co-Boosting [23]	Co-Boosting++
CNN	P	-	33.80±0.45	32.71±1.53	34.25±2.12	35.68±1.83	50.77±1.78	**53.18±0.87**
	A	-	22.56±0.43	21.17±0.52	21.14±0.78	22.41±0.39	23.54±1.18	**26.29±1.31**
	C	-	29.65±0.30	29.32±0.83	29.61±0.86	29.01±0.99	37.59±1.41	**40.61±0.94**
	S	-	30.45±0.49	29.85±0.26	28.36±0.76	30.17±0.34	30.67±2.67	**32.25±1.30**
	AVG	-	29.12±0.42	28.26±0.79	28.34±1.13	29.34±0.89	35.64±1.66	**37.91±1.08**
ResNet	P	11.32±1.93	37.09±0.45	36.50±1.04	37.55±1.78	37.19±1.89	51.43±1.72	**54.42±1.41**
	A	18.50±2.88	23.76±0.85	22.95±0.82	22.92±1.89	24.83±1.49	26.76±1.23	**29.93±0.95**
	C	16.60±1.28	29.62±0.48	28.53±0.89	29.35±1.68	31.78±1.30	36.73±1.24	**39.82±1.40**
	S	19.65±2.63	30.20±0.42	29.77±0.85	29.67±0.76	32.00±1.20	35.35±1.74	**37.55±1.10**
	AVG	16.52±2.18	30.16±0.55	29.31±0.90	29.87±1.53	31.45±1.47	37.56±1.48	**40.97±0.83**

TABLE IX
TEST ACCURACY OF THE TARGET SERVER MODEL OF DIFFERENT METHODS WHEN COMBINING ADVANCED LOCAL TRAINING

Dataset	FedAvg	FedDF	F-ADI	F-DAFL	DENSE	Co-Boosting [23]	Co-Boosting++
SVHN	43.46±3.18	62.67±1.12	56.94±1.35	61.18±1.59	62.00±1.47	68.34±1.39	**71.64±0.93**
CIFAR-10	23.39±3.93	39.54±1.69	37.88±1.83	38.53±0.60	41.14±1.65	49.04±0.73	**51.85±0.91**

TABLE X
TEST ACCURACY OF THE ENSEMBLE (TEACHER) IN THE VARYING LOCAL DATA AMOUNT SETTING

Method	σ=0.4	σ=0.8	σ=1.2
FedENS	46.87±1.02	41.86±1.20	37.88±1.38
DW-FedENS	47.80±1.21	53.25±0.52	47.52±0.40
Co-Boosting [23]	58.94±0.50	57.41±1.12	55.27±1.72
Co-Boosting++	**61.05±1.13**	**59.01±0.92**	**57.49±1.01**

that introducing SAM-based techniques during the local training phase, as compared to the baseline in Table I, indeed enhances the performance of various methods. This improvement is attributed to the potential alignment achieved during local training, which ultimately leads to better server performance upon aggregation. Moreover, our proposed method, Co-Boosting, continues to outperform all the baselines, highlighting the superiority of our approach. Moreover, this shows that our method is not dependent on the specifics of the local training phase. With a better local training method, our approach can further improve.

6) Different Local Data Amount: Our proposed Co-Boosting++ framework integrates an enhanced ensemble strategy with the Mixture of Experts (MoE) architecture. Its key feature is the dynamic adjustment of expert weights based on input data, which enables efficient specialization of experts, balances their contributions, and effectively addresses model heterogeneity. We evaluate its performance through experiments in an unbalanced local data setting. Similar to [42], we sample data amounts for each client from a lognormal distribution. Higher values of σ result in more unequal data distribution. Table X and Fig. 5 display the performance of the ensemble and the server, where the prefix 'DW-' signifies weighted averaging based on the local data amount. As observed, reweighting clients based on their data amount yields some benefits, but it falls short of achieving the best ensemble. Furthermore, when using the

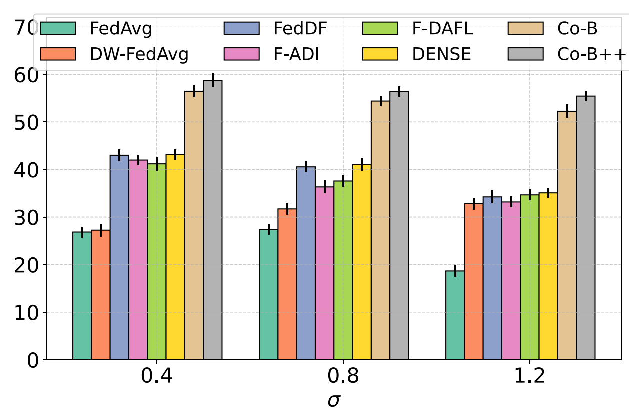

Fig. 5. Test accuracy of the target server model in the varying local data amount setting.

averaged ensemble FedENS as the teacher, all baseline methods perform poorly due to the suboptimal teacher. In contrast, benefiting from simultaneously boosting data and ensemble, the ensemble we get consistently outperforms FedENS. This leads to a substantial performance gain of the target server model achieved in Co-Boosting++ over all baselines, with a margin of at least 10% .

7) Convergence Analysis: To validate the convergence stability and scenario adaptability of the proposed method during optimization, we conduct experiments on three representative datasets, FMNIST, SVHN, and CIFAR-10, with non-IID data partitions of varying degrees, and plot the loss variation curves throughout the training process, as in Fig. 6. As observed from the overall trends, under all three datasets and their respective non-IID partition scenarios, the loss of the proposed method consistently exhibits a pattern of gradual decrease followed by stabilization with increasing training epochs. In the early training stage, the loss drops rapidly to optimize basic classification

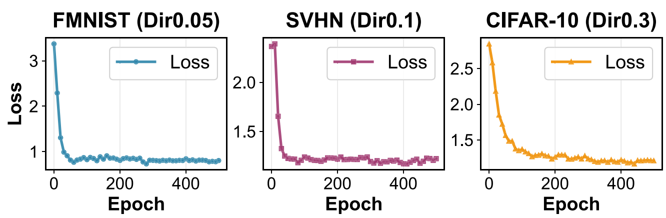

Fig. 6. Loss convergence curves of Co-Boosting++.

errors; in the later stage, it stabilizes within a low-loss range without significant fluctuations.

Improvement over ICLR version [23]: Co-Boosting++ introduces several crucial advancements over the original Co-Boosting framework, resulting in substantial performance gains. Key improvements include the integration of a dynamic Mixture of Experts (MoE) ensemble and the introduction of a diversity-enhanced loss function for synthetic data generation. These innovations work together to provide a more flexible and robust learning process, particularly in challenging non-IID and heterogeneous settings. Additionally, we now simultaneously generate models with different architectures, specifically tailored to meet the requirements of diverse devices. This extension increases both the adaptability and scalability of the generated models, allowing them to perform optimally across a wide range of environments. Experimental results consistently show that Co-Boosting++ outperforms its predecessor, achieving accuracy improvements of over 2% . These enhancements demonstrate the power of mutual reinforcement between data generation and ensemble learning, leading to significant improvements in the final server model.

More discussions: Viewing OFL through the lens of today's pre-trained foundation model era, as highlighted in works like [63], [64] opens up new avenues of exploration. Our work positions itself as a pioneering effort in adapting OFL to the era of foundation models. The key strength of our method is its independence from alterations in local training and its architecture-agnostic nature. This flexibility is particularly relevant in the context of foundation models, which are becoming increasingly central in various domains.

V. Conclusion

In this paper, we seek to tackle the inherent bottleneck of one-shot federated learning, where the performance of the server model is inextricably linked with the quality of the generated data and the ensemble. We propose Co-Boosting++, a novel framework that iteratively enhances both components through a mutual optimization process. By generating hard samples in an adversarial manner and dynamically refining the ensemble via a Mixture of Experts (MoE) framework, Co-Boosting++ effectively captures diverse knowledge from heterogeneous pre-trained models. Furthermore, we extend the framework to support the generation of multiple heterogeneous target models, ensuring adaptability to diverse deployment constraints. Extensive experiments demonstrate that Co-Boosting++ consistently outperforms existing methods across various settings, highlighting its effectiveness in both homogeneous and heterogeneous model scenarios. Additionally, its practicality in real-world model market applications is reinforced by its ability to operate without local training modifications or additional data/model transmissions.

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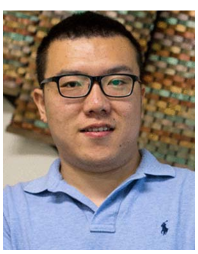

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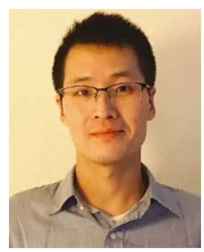

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